Computer Networks – Solution Scheme for End Semester

SECTION -A

1-(a) ICMP belongs to the Network layer of the TCP/IP model.
It works alongside IP and is mainly used for sending control messages, such as reporting errors or checking network reachability. Examples include ping (echo request/reply) and destination-unreachable messages.
ICMP cannot be used for data transmission. It is not designed to carry application data. Its purpose is diagnostic and control communication. ICMP packets do not provide reliability, sequencing, or port numbers, and routers often limit or block them for security reasons. So it cannot replace TCP or UDP for sending real user data.
(b) NAT is important for IPv4 networks because IPv4 has a limited number of public IP addresses. NAT helps extend the life of IPv4 by allowing many devices inside a private network to share a single public IP.

(c) In TCP, the acknowledgment number tells you the next byte the receiver expects.
So if the acknowledgment number is 1501, it means the receiver has received bytes 1000 through 1500.

Data carried = last received byte − first sequence number + 1

= 1500 − 1000 + 1

= 501 bytes

But TCP counts the payload size directly as:

Ack number − Sequence number = 1501 − 1000 = 501

So the TCP segment carried 501 bytes of data.

(d) Slotted ALOHA improves performance by reducing the chances of collisions.

In Pure ALOHA, a station can transmit at any time. If two frames overlap even slightly, both are lost. This leads to a high collision rate.

Slotted ALOHA divides time into fixed-length slots, and stations are allowed to transmit only at the start of a slot. Because transmissions are aligned, collisions happen only when two stations choose the same slot.

(e) The four main steps in the DHCP process are:

1. DHCP Discover

2. DHCP Offer

3. DHCP Request

4.DHCP Acknowledgment (DHCP ACK)

(f) With non-persistent HTTP, the client opens one TCP connection for each object.

So, if the client requests 5 objects, the number of TCP connections opened is:

5 TCP connections

(g) The preamble in an Ethernet frame helps the receiver get ready to read the incoming data.

It is a sequence of bits (typically 7 bytes of alternating 1s and 0s) that lets the receiver synchronize its clock with the sender. Once the clocks are aligned, the receiver can correctly interpret the rest of the frame.

(h) DNS queries usually use UDP on port 53.

(i) The window size is 4000 bytes and each segment can carry 1000 bytes (MSS).

Number of segments that fit in the window:

4000 ÷ 1000 = 4 segments

So, the sender can transmit 4 TCP segments before it needs to wait for an acknowledgment.

(j) Packet Switching

- Data is broken into packets.
- Each packet may take a different route to reach the destination.
- No dedicated path is reserved.
- More efficient use of the network because bandwidth is shared.
- Suitable for data communication like internet traffic.

- Delay may vary because of queueing and different paths.

Circuit Switching

- A dedicated communication path is established before transmission.
- The path remains reserved for the entire session.
- Constant and predictable delay.
- Inefficient for bursty data because the reserved line may stay idle.
- Used in traditional telephone networks.

SECTION-B

2(a) Initial window size or window size for first transmission: cwnd = 2 MSS
For 2nd transmission cwnd = 2 ^ 2 = 4MSS

For 3rd transmission → cwnd = 2 ^ 3 = 8MSS

When threshold is reached then additive increase
For 4th transmission cwnd = 8 + 1 = 9 MSS
For 5th transmission cwnd = 9+ 1 = 10 MSS(fails)
Now window size reduced to 10/2 = 5MSS
For 6th transmission cwnd = 1 MSS
For 7rd transmission cwnd = 2 ^ 1 = 2MSS

For 8th transmission → cwnd = 2 ^ 2 = 4MSS

Thersoled reached now additive increase For 9th transmission cwnd = 4 + 1 = 5MSS
For 10th transmission cwnd = 5 + 16 MSS
So, at the end of 10th successful transmission
The congestion window size will be 6 + 1 = 7 MSS

2(b) (b) TCP header fields: diagram and purposes
Header format (20 bytes minimum, options extend size):

Source port (16): Sender’s port.

Destination port (16): Receiver’s port.

Sequence number (32): Byte sequence position of first data byte.

Acknowledgment number (32): Next expected byte (ACK).

Data offset (4): Header length in 32-bit words.

Reserved (3): Zero, future use.

Flags (9): NS,CWR,ECE,URG,ACK,PSH,RST,SYN,FIN control behaviors.

Window size (16): Receiver's buffer space advertisement (flow control).

Checksum (16): Integrity check of header + data + pseudo-header.

Urgent pointer (16): Offset to urgent data end when URG set.

Options (variable): MSS, Window Scale, SACK, Timestamps…

Reliability:

Seq/Ack numbers, Checksum, Retransmissions (via ACK/Timeout), SACK options.

Flow control:

Window size (and Window Scale option).

Connection management:

SYN, FIN, RST flags; 3-way handshake, 4-way termination.

Marking scheme:

Clean field labeling: 2 marks

Correct purpose mapping: 2 marks

Reliability/flow/connection fields identified: 1 mark

Students must draw the diagram and explain each field

3(a)

a) To divide the network 200.1.2.0/24 into 4 subnets, we need to borrow 2 bits from the host portion since $2^2 = 4$.

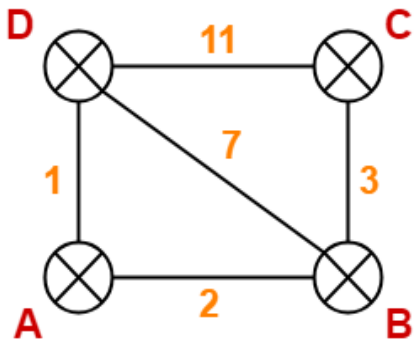
b) As 2 bits are borrowed from the host portion, this changes the subnet mask from /24

(255.255.255.0) to /26 (255.255.255.192).

c) Each subnet has 64 addresses, of which 62 are usable. The network ID, broadcast address, and usable IP range for each of the 4 subsets is as follows.

Subnet	Network ID	Broadcast address	Usable IP range
1	200.1.2.0	200.1.2.63	200.1.2.1 – 200.1.2.62
2	200.1.2.64	200.1.2.127	200.1.2.65 – 200.1.2.126
3	200.1.2.128	200.1.2.191	200.1.2.129 – 200.1.2.190
4	200.1.2.192	200.1.2.255	200.1.2.193 – 200.1.2.254

3(b)



At Router A-

Destination	Distance	Next Hop
A	0	A
B	2	B
C	∞	–
D	1	D

At Router B-

Destination	Distance	Next Hop
A	2	A
B	0	B
C	3	C
D	7	D

At Router C-

Destination	Distance	Next Hop
A	∞	–
B	3	B
C	0	C
D	11	D

At Router D-

Destination	Distance	Next Hop
A	1	A
B	7	B
C	11	C
D	0	D

Each router exchanges its distance vector obtained in Step-01 with its neighbors.
After exchanging the distance vectors, each router prepares a new routing table

Router A receives distance vectors from its neighbors B and D.
Router A prepares a new routing table as-

From B

2
0
3
7

Cost(A→B) = 2

From D

1
7
11
0

Cost(A→D) = 1

Destination	Distance	Next hop
A	0	A
B		
C		
D		

New Routing Table at Router A

Cost of reaching destination B from router A = min { 2+0 , 1+7 } = 2 via B.
Cost of reaching destination C from router A = min { 2+3 , 1+11 } = 5 via B.
Cost of reaching destination D from router A = min { 2+7 , 1+0 } = 1 via D.

Thus, the new routing table at router A is-

Destination	Distance	Next Hop
A	0	A
B	2	B
C	5	B
D	1	D

Router B receives distance vectors from its neighbors A, C and D.
Router B prepares a new routing table as-

From A

0
2
∞
1

Cost (B→A) = 2

From C

∞
3
0
11

Cost (B→C) = 3

From D

1
7
11
0

Cost (B→D) = 7

Destination	Distance	Next hop
A		
B	0	B
C		
D		

New Routing Table at Router B

Cost of reaching destination A from router B = min { 2+0 , 3+∞ , 7+1 } = 2 via A.
Cost of reaching destination C from router B = min { 2+∞ , 3+0 , 7+11 } = 3 via C.
Cost of reaching destination D from router B = min { 2+1 , 3+11 , 7+0 } = 3 via A

Thus, the new routing table at router B is-

Destination	Distance	Next Hop
A	2	A
B	0	B
C	3	C
D	3	A

Like this students must complete for C and D for first round then so on till no changes .

4(a)Translate to binary

Generator x3+1x^3+1x3+1 → binary 1001 (degree 3)

Append 3 zeros to the message (degree = 3):
augmented message = 1010000 000 = 1010000000.
We now perform modulo-2 (XOR) long division of 1010000000 by 1001. (window = 4 bits, because generator length = 4).

Division steps (window before XOR → result after XOR → then drop leading bit and bring next down):

1.Take first 4 bits: 1010

XOR with 1001 → 0011 → drop leading 0 → 011, bring down next bit (0) → 0110

2.Current 4-bit window: 0110 (MSB 0)

XOR with 0000 → 0110 → drop leading 0 → 110, bring down next bit (0) → 1100

3.Current window: 1100

XOR with 1001 → 0101 → drop leading 0 → 101, bring down next bit (0) → 1010

4.Current window: 1010

XOR with 1001 → 0011 → drop leading 0 → 011, bring down next bit (0) → 0110

5.Current window: 0110 (MSB 0)

XOR with 0000 → 0110 → drop leading 0 → 110, bring down next bit (0) → 1100

6.Current window: 1100

XOR with 1001 → 0101 → drop leading 0 → 101, bring down next bit (0) → 1010

7.Current window: 1010

XOR with 1001 → 0011 → now no more bits to bring down; drop leading 0 → 011

8.The final remainder (degree < 3) is 011.
CRC bits = 011.

Transmitted codeword = original message + CRC = 1010000 011 → 1010000011.

Quick check: dividing the full codeword 1010000011 by generator 1001 gives remainder 000 (no error), confirming correctness.

4(b)Slotted ALOHA throughput with 200-bit frames on 200 kbps

Slot time: Frame time =200/200,000=0.001 s → 1000 slots/s.

The frame transmission time is 200/200 kbps or 1 ms.

a) If the system creates 1000 frames per second, or 1 frame per millisecond, then G = 1. So S = G × e −G = 0.368 (36.8 percent). This means that the throughput is 1000 × 0.0368 = 368 frames. Only 368 out of 1000 frames will probably survive. Note that this is the maximum throughput case, percentage-wise.
b) Here G is 1/2. In this case S = G × e −G = 0.303 (30.3 percent). This means that the throughput is 500 × 0.0303 = 151. Only 151 frames out of 500 will probably survive.
c) Now G is 1/4. In this case S = G × e −G = 0.195 (19.5 percent). This means that the throughput is 250 × 0.195 = 49. Only 49 frames out of 250 will probably survive.

5(a)

Distance: 40,000 km = 4.0×10^7 m

Propagation speed: 2.5×10^8 m/s

Rate: $R = 2$ Mbps

a) Propagation delay: $\frac{4.0 \times 10^7}{2.5 \times 10^8} = 0.16$ s

Bandwidth–delay product: $2 \times 10^6 \cdot 0.16 = 3.2 \times 10^5$ bits = 320,000 bits.

b) Max bits in flight (continuous send): 320,000 bits.

c) Bit width: $\frac{\text{prop speed}}{R} = \frac{2.5 \times 10^8}{2 \times 10^6} = 125$ m per bit.

5(b)OSI model functions and TCP/IP comparison

OSI layers:

Physical: Bits on medium; signaling, modulation.

Data link: Framing, MAC, error detection; Ethernet, Wi-Fi MAC.

Network: Logical addressing, routing; IP.

Transport: End-to-end delivery, segmentation, reliability; TCP/UDP.

Session: Dialog control, checkpoints.

Presentation: Syntax, encoding, compression, encryption.

Application: Network services to apps; HTTP, SMTP, FTP, DNS.

TCP/IP model :

Link (Network interface): Physical + Data link equivalents.

Internet: IP routing and ICMP, ARP/ND adjuncts.

Transport: TCP/UDP.

Application: All protocols (HTTP, SMTP, DNS, etc.) including some OSI session/presentation functions.

Similarities/Differences:

Similarities: Both layered; separation of concerns; transport and network concepts align.

Differences: OSI is prescriptive with distinct session/presentation; TCP/IP is pragmatic—these functions live within applications; protocol assignments differ (ICMP with Internet layer).

Implementation reality: TCP/IP dominates deployments; OSI used for conceptual teaching and design clarity.

6(a)Webmail (Alice) to POP3 retrieval (Bob): protocol sequence [5]
Step 1 (Alice to webmail server):

Protocol: HTTPS (HTTP over TLS). Alice’s browser uploads the composed email to the provider’s application backend.

Step 2 (Webmail server to Bob’s mail server):

Protocol: SMTP (port 25/587/Submission). The provider's outbound MTA forwards the message, possibly via intermediate relays, to Bob's domain's MX server.

Step 3 (Bob's server stores mail):

Protocol: SMTP reception; message queued in Bob's mailbox on the mail server.

Step 4 (Bob retrieves mail):

Protocol: POP3 (port 110/995 for POP3S). Bob's mail client connects to his server, authenticates, downloads the message (alternatively IMAP for sync, but question specifies POP3).

Supporting layers: DNS lookups (MX/A records) occur during SMTP routing; TCP underlies HTTPS/SMTP/POP3.

6(b)

(b) Number of IP datagrams for 5,000,000-byte MP3 with 1,500-byte MTU and 20-byte IP header [5]

- **Max IP payload per datagram:** $1500 - 20 = 1480$ bytes.
- **Datagrams required:** $[5{,}000{,}000/1480]$.
- **Computation:**
  - $1480 \times 3378 = 4{,}999{,}440$ bytes; remainder = 560 bytes.
  - One more datagram for remainder → **3379 datagrams** total.